

JACK THOMAS WARFIELD

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PROFESSIONAL APPOINTMENTS

University of California, Berkeley

Postdoctoral Scholar, Department of Astronomy

Supervisor: Daniel Weisz

August 2026 – Present

EDUCATION

The University of Virginia

Ph.D. in Astronomy

May 2026

Dissertation Title—*Beyond the Milky Way: Space-Based Proper Motion Measurements in the Era of JWST*

Advisor: Nitya Kallivayalil

M.S. in Astronomy

May 2022

The Ohio State University

B.S. in Physics and Astronomy & Astrophysics

May 2020

with honors research distinction

Thesis Title—*An Intermediate-Age α -Rich Galactic Population*

Beyond the Solar Neighborhood

Thesis Advisors: Marc Pinsonneault, Jennifer Johnson, Joel Zinn

PUBLICATIONS ([ADS Query](#))

ORCID: [0000-0003-1634-4644](https://orcid.org/0000-0003-1634-4644)

25 total, 5 first-author, h-index: 13 (NASA ADS, October 2025)

25. J. Sánchez Almeida, A. R. Plastino, S. Guerra Arencibia, et al., *Constraining the shape of dark matter haloes using only starlight II. Tests of the technique with objects of known gravitational potential*, [arXiv:2607.27001](#), accepted to A&A, 2026.
24. S. T. Sohn, P. Bennet, K. A. McKinnon, et al., *Hubble Astrometry for the Local Group and Beyond in the 2030s*, [arXiv:2605.24191](#), STScI and NASA GSFC “Building a Roadmap for Hubble Science into the 2030s” white paper, 2026.
23. C. M. Slaughter, E. D. Skillman, A. Savino, et al., *The JWST Resolved Stellar Populations Early Release Science Program. IX. The RR Lyrae Population in WLM with HST and JWST*, [ApJ 1006, 253S](#), 2026.
22. J. T. Warfield, K. A. McKinnon, S. T. Sohn, et al., *The Proper Motion of Draco II with HST using Multiple Reference Frames and Methodologies*, [ApJ 998, 3W](#), 2026.
21. N. Ahvazi, A. B. Pace, C. T. Garling, et al., *The abundance and properties of the lowest luminosity dwarf galaxies around the Milky Way: Insights from Semi-Analytic Models*, [arXiv:2511.15808](#), submitted to OJAp, 2025.
20. C. T. Garling, N. Kallivayalil, K. B. W. McQuinn, et al., *Measuring Resolved Star Formation Histories from High-Precision Color-Magnitude Diagrams with StarFormationHistories.jl*, [ApJS 277, 61G](#), 2025.
19. R. E. Cohen, K. B. W. McQuinn, A. Savino, et al., *The JWST Resolved Stellar Populations Early Release Science Program. VIII. The Spatially Resolved Star Formation History of WLM*, [ApJ 981, 153C](#), 2025.

18. A. Savino, M. Gennaro, A. E. Dolphin, et al., *The JWST Resolved Stellar Populations Early Release Science Program. VII. Stress Testing the NIRCам Exposure Time Calculator*, [ApJ 970, 36S, 2024](#).
17. M. L. Boyer, G. Pastorelli, L. Girardi, et al., *The JWST Resolved Stellar Populations Early Release Science Program. VI. Identifying Evolved Stars in Nearby Galaxies*, [ApJ 973, 120B, 2024](#).
16. H. Richstein, N. Kallivayalil, J. Simon, et al., *Deep Hubble Space Telescope Photometry of LMC and Milky Way Ultra-Faint Dwarfs: A careful look into the magnitude-size relation*, [ApJ 967, 72R, 2024](#).
15. D. R. Weisz, A. E. Dolphin, A. Savino, et al., *The JWST Resolved Stellar Populations Early Release Science Program. V. DOLPHOT Stellar Photometry for NIRCам and NIRISS*, [ApJS 271, 47W, 2024](#).
14. J. T. Warfield, J. C. Zinn, J. Schonhut-Stasik, et al., *The APO-K2 Catalog. II. Accurate Stellar Ages for Red Giant Branch Stars Across the Milky Way*, [ApJ 967, 72R, 2024](#).
13. J. Schonhut-Stasik, J. C. Zinn, K. G. Stassun, et al., *The APO-K2 Catalog. I. 7,673 Red Giants with Fundamental Stellar Parameters from APOGEE DR17 Spectroscopy and K2-GAP Asteroseismology*, [AJ 167, 50, 2024](#).
12. K. McQuinn, M. Newman, A. Savino, et al., *The JWST Resolved Stellar Populations Early Release Science Program. IV. The Star Formation History of the Local Group Galaxy WLM*, [ApJ 961, 16, 2024](#).
11. C. Jennings, A. Savino, D. Weisz, et al., *The Hubble Space Telescope Survey of M31 Satellite Galaxies. III. Calibrating the Horizontal Branch as an Age Indicator for Nearby Galaxies*, [arXiv:2311.16397](#), Submitted to AJ, 2023.
10. J. J. Han, A. Dey, A. M. Price-Whelan, et al., *NANCY: Next-generation All-sky Near-infrared Community survey*, [arXiv:2306.11784](#), Roman Core Community Survey white paper, 2023.
9. J. T. Warfield, H. Richstein, N. Kallivayalil, et al., *The JWST Resolved Stellar Populations Early Release Science Program. III. Photometric Star-Galaxy Separations for NIRCам*, [Res. Notes AAS 7, 2, 2023](#).
8. D. Weisz, K. McQuinn, A. Savino, et al., *The JWST Resolved Stellar Populations Early Release Science Program II. Survey Overview*, [ApJS 268, 15W, 2023](#).
7. J. T. Warfield, N. Kallivayalil, P. Zivick, et al., *HubPUG: Proper Motions for Local Group Dwarfs observed with HST utilizing Gaia as a Reference Frame*, [MNRAS, 519, 1189, 2023](#).
6. H. Richstein, E. Patel, N. Kallivayalil, et al., *Structural parameters and possible association of the Ultra-Faint Dwarfs Pegasus III and Pisces II from deep Hubble Space Telescope photometry*, [ApJ, 933, 217, 2022](#).
5. Abdurro'uf, K. Accetta, C. Aerts, et al., *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar and APOGEE-2 Data*, [ApJS 259, 35, 2022](#).
4. J. C. Zinn, D. Stello, Y. Elsworth, et al., *The K2 Galactic Archaeology Program Data Release 3: Age-abundance patterns in C1-C8, C10-C18*, [ApJ 926, 191, 2022](#).
3. J. T. Warfield, J. C. Zinn, M. H. Pinsonneault, J. A. Johnson, D. Stello, Y. Elsworth, R. A. García, T. Kallinger, S. Mathur, B. Mosser, R. L. Beaton, and D. A. García-Hernández, *An Intermediate Age Alpha-Rich Galactic Population in K2*, [AJ 161, 100, 2021](#).

2. J. C. Zinn, D. Stello, Y. Elsworth, et al., *The K2 Galactic Archaeology Program Data Release 2: Asteroseismic Results from Campaigns 4, 6, & 7*, *ApJS* **251**, 23, 2020.
1. M. S. Haseman, P. Saadaktia, **J. T. Warfield**, J. Lawrence, A. Harnandez, G. E. Jellison, L.A. Boatner, and F. A. Selim, *Optical and electrical properties of Sn-doped ZnO single crystals*, *Journal of Electronic Materials* **47**, 1497, 2018.

OBSERVING PROGRAMS

7. N. Kallivayalil, S. T. Sohn, A. Wetzel, et al.; *Dynamically Mapping the Satellite Galaxies in the Outer Halo of the Milky Way*; *HST Cycle 33*, ID. #18068.
6. P. Bennet, E. Patel, et al.; *The Mysterious Death of a Lonely Dwarf: Who Quenched Cetus?*; *HST Cycle 33*, ID. #18032.
5. P. Bennet, E. Patel, et al.; *Proper Motions of the M31 Satellites: The missing pieces of the 6D puzzle*; *HST Cycle 33*, ID. #18029.
4. P. Bennet, R. van der Marel, et al.; *Enabling cross instrument proper motions with Draco dSph and NGC 2419*; *HST Cycle 32*, ID. #17926 & *JWST Cycle 3*, ID. #9225.
3. P. Bennet, R. van der Marel, et al.; *20 Years of time baseline, The 3D kinematics of Centaurus A*; *HST Cycle 32*, ID. #17925.
2. **J. T. Warfield**, N. Kallivayalil, H. Richstein, et al.; *Moving beyond the Milky Way: Enabling Cross-Observatory Proper Motion Determinations with HST and JWST*; *HST Cycle 31*, ID. #17473 & *JWST Cycle 2*, ID. #4570.
1. L. Strigari, W. Cerny, A. Drlica-Wagner, et al.; *Creating a Standardized Sample of Milky Way Satellite Galaxy Binary Fraction Measurements*; *HST Cycle 30*, ID. #17056.

AWARDS & HONORS

Jefferson Scholars Foundation Dissertation Year Fellowship	2025
Emma Williams Prize: outstanding research of a 4th year graduate student	2024
The Raven Society (Inductee)	2023
L. Earl Slusher Scholarship in Astronomy	2019
Summer Undergraduate Research Program Astronomy Undergraduate Scholarship	2019
Honors & Scholars Enrichment Grant	2017

TEACHING EXPERIENCE

Course Grader <i>The University of Virginia</i> <i>Courses: Astronomy 1210</i> Grading for undergraduate non-major general education course.	<i>Aug. 2024 - Dec. 2024</i>
Course Grader <i>The University of Virginia</i> <i>Courses: Astronomy 5450 & Physics 5640</i> Grading for graduate level courses on high-energy astrophysics and computation physics.	<i>Jan. 2023 - May 2023</i>

Graduate Teaching Assistant*Aug. 2020 - Dec. 2020**The University of Virginia**Course: Astronomy 1230*

Oversaw night labs and grading for undergraduate non-major observational astronomy course.

Undergraduate Teaching Assistant*Jan. 2020 - May 2020**The Ohio State University Department of Astronomy**Courses: Astronomy 1221, 5681*

Lecture assistant, tutor, and grader for Astronomy/Physics major introductory data analysis and programming course. Undergraduate teaching assistant and grader for upper-division major stellar structure and nucleosynthesis course.

Undergraduate Teaching Assistant*Aug. 2019 - Dec. 2019**The Ohio State University Department of Astronomy**Course: Astronomy 3350*

Lecture assistant, tutor, and grader for Astronomy major data analysis and programming course.

Physics Tutor*Sept. 2017 - May 2019**The Ohio State University Department of Physics**Courses: Physics 1201, 1202, 1251, 1252*

Open tutoring for introductory non-major and major physics courses.

Instructional Assistant*Jan. 2017 - May 2017**The Ohio State University Department of Astronomy**Course: Astronomy 1140*

Prepared materials, assisted in lecture, and graded assignments for a general education astronomy course.

Supplemental Instructor*Jan. 2016 - May 2016**Bowling Green State University Learning Commons**Course: Physics 2010*

Designed and ran 3 weekly sessions to go over material being covered in an introductory non-major physics course.

PRESENTATIONS

Galactic Frontiers II: Dwarf Galaxies in the Local Volume and Beyond

*June 2025**Beyond the Milky Way: Proper Motion Measurements in the Era of JWST*

The Colonnades Senior Living

*Oct. 2024**Out of Sight, Front of Mind: The Puzzle of Dark Matter*

JHU/STScI Galaxy Journal Club

*May 2024**The Milky Way and Beyond: Probing Near-Field Cosmology from Proper Motion Measurements in the Local Group*

Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond

*July 2023**HubPUG: Utilizing Gaia as a Reference Frame to Measure Proper Motions for Local Group Dwarfs with HST*

AAS 241

*Jan. 2023**HUBPUG: Combining HST and Gaia data to Measure Proper Motions for Local Group Dwarfs*

SDSS-IV & V Collaboration Meeting

June 2020

Revealing an Intermediate Age Population of Alpha-Rich Red Giants in the K2 Fields

AAS 235

Jan. 2020

Revealing an Intermediate Age Population of Alpha-Rich Red Giants in the K2 Fields

OUTREACH

Dark Skies Bright Kids (DSBK)

Member

Oct. 2020 - Present

DSBK is an astronomy outreach organization primarily run by graduate students at the University of Virginia seeking to enhance science education in Virginia elementary schools. We run programs for students in the Charlottesville area during the academic year (including weekly after-school astronomy clubs) as well as week-long summer camps for students across Virginia.

The Astronomical Society at The Ohio State University

Member

Aug. 2016 - July 2020

Secretary

May 2017 - May 2019

President

May 2019 - July 2020

Organized roof nights and astronomy talks for undergraduate and community audiences. Organized a welcome event for incoming undergraduate Astronomy students for the Department of Astronomy. Worked with other student organizations and university departments to organize and secure funding for a yearly trip to the Green Bank Observatory for undergraduate astronomy majors.

Polaris

Mentor

Aug. 2019 - May 2020

Polaris offers mentoring for first- and second-year undergraduate physics and astronomy majors by graduate and senior undergraduate students. I worked individually with a freshman student to help them navigate through their first year at Ohio State and find opportunities to best achieve their goals. In the spring 2020 semester, I guided them through a small research project in astronomy working with APOGEE and Kepler asteroseismology data.

TECHNICAL SKILLS

Programming Languages

Python, C++, Wolfram Language (Mathematica), Julia, bash

Markup Languages

L^AT_EX, HTML, CSS, PHP